Agentic AI Trading System — Project Overview

Objective

We want to build an agentic AI system that screens a predefined universe of equities (single names, sector ETFs, and country ETFs) using specialized agents that conduct macro research, fundamental research, technical research, risk management, and portfolio management. The system would generate trade recommendations on a daily, weekly, or monthly basis that we'd then evaluate on a discretionary basis. We'll provide the agents with a defined set of risk management guidelines and a long/short investment process they must adhere to, so that every recommendation fits the constraints of our strategy. We will need to run this on a personal device outside of the Graham Capital ecosystem but would like a way to safely interact with the agents while at the desk.

Universe

All single-name equities in the S&P 500, the 11 GICS sector ETFs (e.g., XLE for energy), 20 predefined MSCI country ETFs, plus gold and gold miners, silver and silver miners, copper, aluminum, and oil and gas. The agents should be working continuously to identify opportunities across this universe.

Investment Process

Agents pitch trade ideas whenever they identify an opportunity. Trades can be long or short at the single-name, sector ETF, or country ETF level. Because of our risk management constraints, every bet has to be hedged. If the energy agent wants to go long a single name in energy, it also has to short the equal-weight ETF for that sector … in this case RSPG. If an agent finds an opportunity (long or short) at the sector ETF level, it pairs the trade with an opposite-direction position in an equal-weight S&P 500 ETF in this case RSP. At the country level, a developed-market idea gets paired with an opposite-direction trade in a DM ETF or ACWI, whichever it sees fit; an emerging-market idea gets paired with an EM ETF like EEM or ACWI. Agents don't need to always have a position on, if nothing looks attractive, an agent can stay in cash. Fundamental and Macro agents will be the agents who are debating and typically generating trade ideas. These ideas will be evaluated by the technical agent(s) and then by the risk management agent(s). Technical agents can also find opportunities, but they must be debated with the fundamental or macro analysts and always approved by risk management. The portfolio manager agent will ultimately determine if the trade should be placed in the book/sent to us.

Risk Management Metrics

Target returns of 12–15%+ annualized, with 6% downside volatility. Each position should carry a 2–3% stop loss and a 5% gain target. The book tracks 12-month factor betas to: DXY, SPX, USGG10Yr, VIX, Growth/Value, CL1, Large/Small cap, CDX HY 5Y, and XAU. The combined book should never run a factor beta beyond ±0.6 to any individual factor.

Fundamental Agents

Eleven fundamental equity agents, each a sector specialist for one of the GICS sectors in the S&P 500, running bottom-up fundamental research at both the single-name and sector level. The work should look like what a long/short equity analyst does: build the model, work the financials … revenue drivers, margins, free cash flow, balance sheet and leverage … and value names on the multiples and DCF that actually matter, benchmarked against peers and their own history. The goal is variant perception: finding where our numbers and thesis diverge from consensus, on the long side (mispriced quality, underappreciated catalysts, positive estimate revisions) and the short side (stretched valuations, deteriorating fundamentals, structural decline). Critically, the analysis has to be sector specific, each agent works the KPIs and drivers that matter for its space (production, reserves, and breakevens in energy; NIM, credit quality, and capital ratios in financials; net revenue retention and ARR in software; backlog and book-to-bill in industrials) rather than running a generic screen. These agents debate with each other to identify opportunities, either in individual equities or at the sector level expressed through that sector's ETF. They can use news sources, chat rooms, whatever they need to get a pulse on the market and find opportunities, but everything ultimately has to align with their fundamental view.

Macro Agents

These agents are regional or product specific: a LatAm macro agent, plus CEEMEA, Asia, North America, Western Europe, and a Commodities specialist. The work should look like what a top-down macro analyst does: track growth and activity (GDP, PMIs, labor), inflation and the central bank reaction function, the fiscal picture and debt sustainability, and the external accounts, current account, balance of payments, reserves, terms of trade, and capital flows … then layer in FX valuation, politics, and policy and geopolitical risk. The point is to form a regional view and map it into the cleanest expression at the country ETF level. Like the equity agents, the analysis has to be region or product specific, each agent owns the drivers that matter for its space (fiscal slippage and commodity terms of trade in LatAm; energy and geopolitics in CEEMEA; China's credit impulse and the trade cycle in Asia) rather than a generic global read. The Commodities specialist works supply/demand balances, inventories, the cost curve, and the shape of the futures curve (contango vs. backwardation) across gold, silver, copper, aluminum, and oil and gas. They debate with each other to identify opportunities at the country ETF and commodity level. Like the fundamental agents, they can use news sources, chat rooms, whatever they need to get a pulse on the market, but everything ultimately has to align with their macro view.

Technical Agent(s)

These agents evaluate the fundamental and macro agents' trade ideas on a technical basis: price momentum, moving averages, volume, option pricing, signals like KST, and anything else a professional technical trader would look at and assign each trade a technical score. They can also surface purely technical opportunities, but those have to be debated with the portfolio manager and the macro and fundamental agents before the PM reviews them. Technical agents are also responsible for monitoring momentum as well as mean reversion indicators to identify sustainable trends. They will also look at CTA levels, supports, resistances, and Fibonacci levels to identify trailing stop losses and take profit levels. They will provide this information to the portfolio manager agent.

Risk Management Agent

Responsible for determining whether the fundamental, macro, and technical agents' trade ideas meet the risk management guidelines we've provided. The risk management agent helps ensure that the factor exposure and idiosyncratic exposure is within limits to make sure we do not exceed overall drawdown limits. They should also help regulate the leverage of the program. When making money and high conviction low correlated trade the leverage should be high and vice versa. This agent should operate as a risk manager at an elite macro hedge fund.

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Portfolio Management Agent

Receives the ideas generated by the fundamental, macro, and technical agents and approved by risk management, then decides whether a trade should go into the book and be sent to me via email or whatever interface we land on. The PM agent is also responsible for dicing the trade sizing, stop losses and adjusting positions to maximize P&L and minimize drawdowns. This agent should operate like a macro portfolio manager at an elite macro hedge fund.